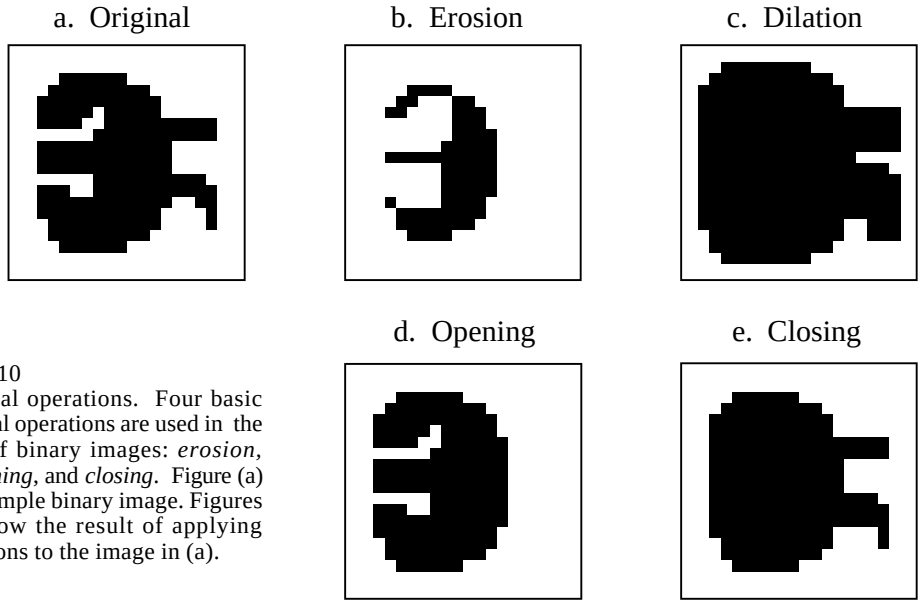

FIGURE 25-10
Morphological operations. Four basic morphological operations are used in the processing of binary images: *erosion*, *dilation*, *opening*, and *closing*. Figure (a) shows an example binary image. Figures (b) to (e) show the result of applying these operations to the image in (a).

other words, each application requires a custom solution developed by trial-and-error. This is usually more of an art than a science. A bag of tricks is used rather than standard algorithms and formal mathematical properties. Here are some examples.

Figure 25-10a shows an example binary image. This might represent an enemy tank in an infrared image, an asteroid in a space photograph, or a suspected tumor in a medical x-ray. Each pixel in the background is displayed as white, while each pixel in the object is displayed as black. Frequently, binary images are formed by thresholding a grayscale image; pixels with a value greater than a threshold are set to 1, while pixels with a value below the threshold are set to 0. It is common for the grayscale image to be processed with linear techniques before the thresholding. For instance, *illumination flattening* (described in Chapter 24) can often improve the quality of the initial binary image.

Figures (b) and (c) show how the image is changed by the two most common morphological operations, **erosion** and **dilation**. In erosion, every object pixel that is touching a background pixel is changed into a background pixel. In dilation, every background pixel that is touching an object pixel is changed into an object pixel. Erosion makes the objects smaller, and can break a single object into multiple objects. Dilation makes the objects larger, and can merge multiple objects into one.

As shown in (d), **opening** is defined as an erosion followed by a dilation. Figure (e) shows the opposite operation of **closing**, defined as a dilation followed by an erosion. As illustrated by these examples, *opening* removes small islands and thin filaments of *object pixels*. Likewise, *closing* removes